

Nationwide Auction System Architectural Document

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I. Kata Overview

An auction company wants to take their auctions online to a nationwide scale. This system is designed to allow users to choose an auction and participate at anywhere in real-time. This system needs to handle large volume of participants, and it offers both live and online auctions. Ideally, the system supports auctions as many simultaneous auctions as possible, and it provides post-auction video streaming. The system manages financial transitions with money exchange available, and it tracks the participate reputations.

II. Problem statement

The current auction company does not have a working online auction system. The goal is to bring a traditional auction company to the next level by moving their operations online on a nationwide scale. Scalability is a firm requirement because the company is expanding aggressively merging with smaller competitors and expects overseas replication. Performance and elasticity are critical because the system is expected to handle hundreds to thousands of participants per auction and to support multiple simultaneous auctions. Availability and reliability are essential because the platform must handle money exchanges, support post-auction video streaming, and provide user reputation checking.

III. Architecturally Interesting Bits

A. Functional Requirements:

1. **User Management:** Ability for users to register, login, and manage their profiles.
2. **Auction Search and Item Listing:** Users must be able to search for the auctions they are interested in. Users can list items for auction, including descriptions, starting bid, and auction duration.
3. **Bidding and Auction Event Processing:** Both live and online auctions happen in real time, where users can place bids on items.
4. **Financial Transaction Processing:** Secure handling of financial transactions when auctions close.
5. **Reputation Checking Processing:** All participants in auctions must be tracked via reputation indexes.

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6. **Video Streaming Processing:** Ability for users to stream video of the auction after the fact.

B. Non-Functional Requirements:

1. **Scalability:** This system needs to handle large volume of participants and supports auctions as many simultaneous auctions as possible.
2. **Data Consistency:** Bidding, money exchange handling and reputation tracking requires data consistency.
3. **Extensibility:** Aggressive expansion and potential replicate overseas.
4. **Performance and Elasticity:** Unpredictable spikes in user and request load.
5. **Availability and Reliability:** With financial transactions and real time auctions, reliability is critical.

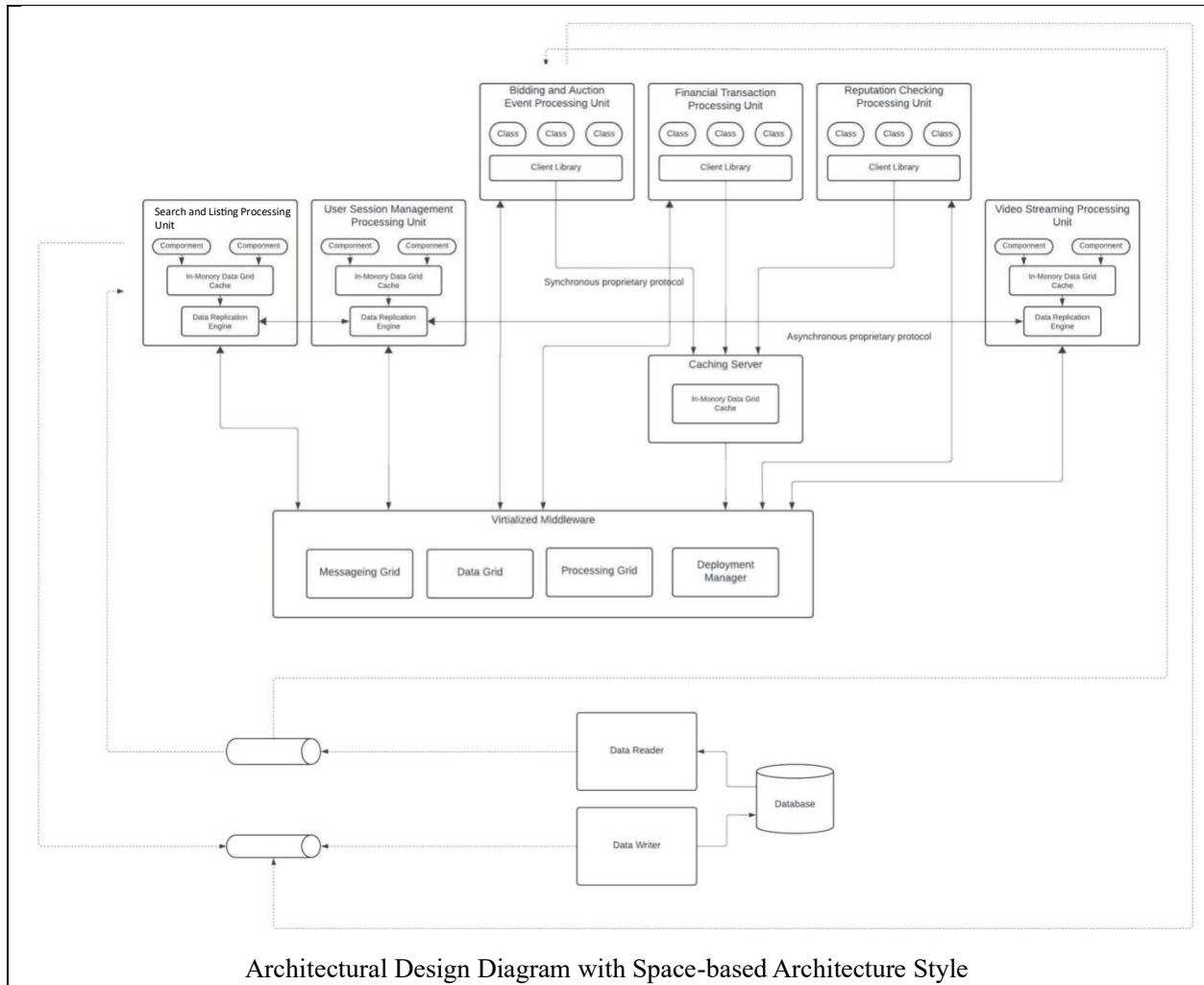
C. Data Storage and Capacity Estimates [2]:

1. **User Data:** Assume the system supports 100,000 active users. Assume each user record takes about 1 KB. $100,000 \text{ users} * 1 \text{ KB} = 100 \text{ MB}$.
2. **Daily Active Users:** Assume up to 10% of the user base is active simultaneously, which is $100,000 \text{ users} * 10\% = 10,000 \text{ users}$.
3. **Auction Data:** Each user lists an average of 5 auctions per month, and each auction record takes about 500 bytes, plus additional data for images and descriptions (let's allocate 100 KB per auction). $500,000 \text{ auctions} * 100.5 \text{ KB} \approx 50 \text{ GB per month}$.
4. **Bid Data:** On average, each auction receives 50 bids. Each bid record takes about 100 bytes. $500,000 \text{ auctions} * 50 \text{ bids} * 100 \text{ bytes} \approx 2.5 \text{ GB per month}$.

IV. Proposed Solution with Space-based Architecture Style

Space-based architecture is well suited for online auction system because it supports high performance, high scalability and high levels of elasticity to handle unpredictable spikes in user and request load.

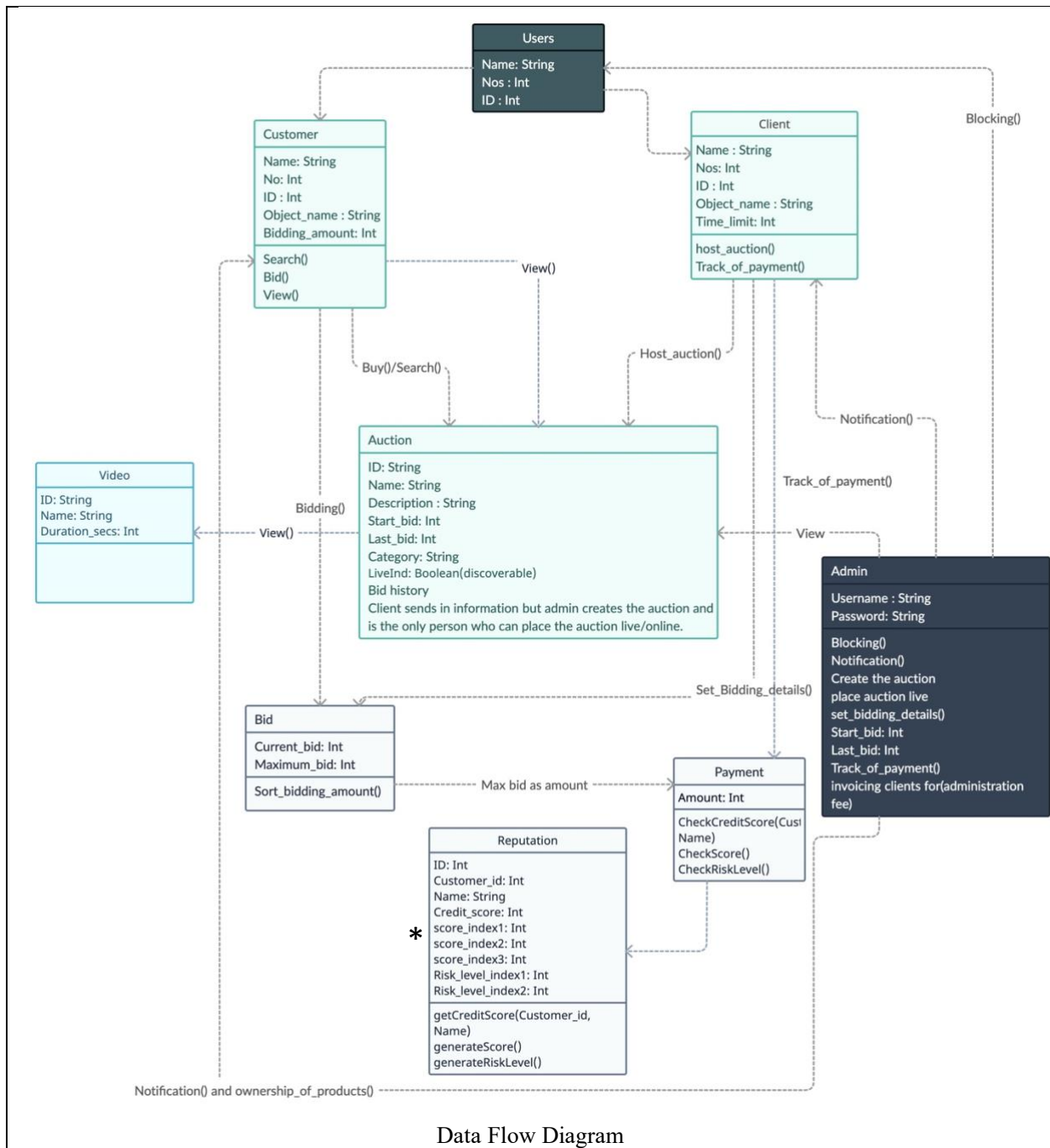
Multiple processing units can be started as the load increases; and as the auction winds down, unused processing units could be destroyed. Individual processing units can be devoted to each auction, ensuring consistency with bidding data. Due to the asynchronous nature of the data pumps, bidding data can be sent to other processing without much latency [1].



Main components dive in:

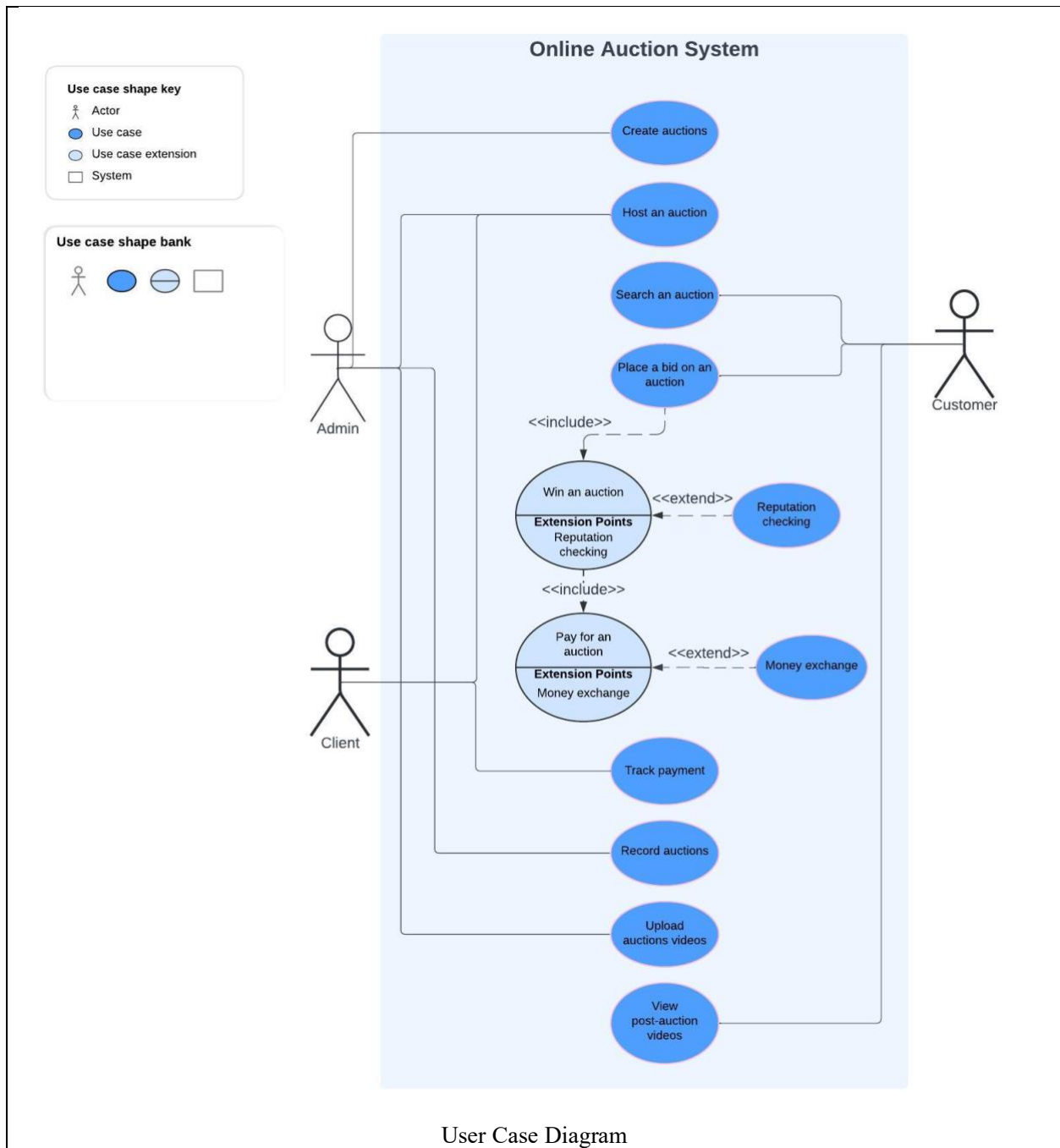
1. **Processing Units:** Processing units are divided by functional. Deploy Bidding and Auction Event, Financial Transaction, and Reputation Checking processing units for each auction, which can be scaled up or down to accommodate the load. Each processing unit contains its own memory and processing power, handling all bidding, participant tracking, and reputation scoring.
2. **Caching Server:** A distributed caching server will store active auction data, real-time bidding updates, financial transactions and participant records that used in user reputation checking. The server enables low-latency data access across nodes and ensures data consistency.
3. **Data Grid:** A replicated data grid will store user data and listing data.
4. **Deployment Manager:** The deployment manager manages the dynamic scale of processing unit instances based on load conditions.

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*score_index1, score_index2, score_index3, and Risk_level_index1, and Risk_level_index2 are machine learning algorithms or methods to determine the user's reputation, such as Decision Tree, Kernel Machines, Clustering, Multilayer Perceptron, Random Forest and Linear Discrimination.

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Personas:

The client: James is an antiques seller, and he wants to expand his reach with an online auction platform that's easy to use and flexible. James wants a streamlined experience for listing and categorizing items, enabling buyers to find his offerings easily. He also seeks transparent, detailed reports on his sales to streamline inventory and accounting. James is looking for a

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system that fosters trust and engagement with new buyers. Above all, he needs reliable tools to interact with participants in real-time, so his auctions can match the energy of a live event.

The customer: Sarah is a passionate collector who loves the excitement of online bidding. For Sarah, discovering the perfect auction is essential; she needs effective search and filtering options to find items by category, location, and price. She likes a real-time bidding experience, secure transparent payment and delivery processes. Sarah's ideal auction platform offers reliability, transparency, and smooth transactions, keeping her engaged and excited to return.

The admin: Alex manages the auction platform, working behind the scenes to ensure each auction runs smoothly and that both clients and customers are supported. From creating auctions to hosting live events, Alex's role is both administrative and customer-facing. They rely on a centralized interface to manage multiple auctions, assist clients and customers, and resolve issues quickly. Alex's focus is on reliability, user satisfaction, and maintaining a smooth, responsive bidding experience that keeps the platform running seamlessly.

V. Solutions that Considered but Discarded

Event-Driven Architecture: Event-driven architecture allows high scalability, performance and fault tolerance. Bidding is an event-based model, it makes me to consider this architecture style at the first place. Since the auctions must be as real-time as possible, handle payment processing and requires reputation tracking, data consistency is essential. Data loss could happen between event propagation, and its lack of elasticity.

Microkernel Architecture: Since the company expanding aggressively by merging with smaller competitors, which means the system needs to have an ability to combine other systems. The core kernel manages the essential auction processes, such as auction searching, bid management, and user registration, and the plugins handles payment processing, video streaming, money exchange, reputation tracking, and integrating with competitors' system after merging. Unfortunately, microkernel architecture doesn't not scale well. Scalability, fault tolerance and elasticity are its weakness.

VI. Deployment Environment

Using a cloud provider capable of autoscaling and deploying multiple instances in various regions to support global reach. Distributed across availability zones to handle regional data and minimize latency. Processing units will be added or removed based on auction demand, triggered by user activity and workload requirements and deployed by deployment manager.

Make sure to set up with multi-region failover to ensure continuity in case of localized issues.

Ensure data backup and redundancy for potential data loss.

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VII. Fitness Function(s)

Canary test	Monitoring critical metrics, e.g., unavailable, failure, response times.
Gatling	Load testing to ensure that the system can handle thousands of participants and simultaneous auctions.
Latency Benchmarking	Use Grafana Dashboard to monitor processing units' traffic, e.g., backend latency, connection latency.
Availability Monitor	Recommend the auction system deploy in any major cloud platform for fault tolerance support.
Consistency checks	Run tests to verify data consistency for financial transactions and reputation checking to ensuring no race conditions or inconsistencies.

VIII. Failure Cases and Mitigation

Failure Cases	Explanation	Mitigation	Possibility	Impact
Node failure	If a processing unit fails, the deployment manager in virtualized middleware space ensures that another processing unit starts up to reducing down time.	Deploy the major part of the system in cloud platforms to be able to auto-scaling and monitor failures.	Medium	Low
Data collisions	Due to replication latency of the caching product.	Reducing the number of processing units, reducing the cache size, or a better caching product.	Low	High
Data inconsistency	If data grid fails, data inconsistency happens between replicas, and it will affect the bidding accuracy.	Use distributed cache instead of replicated cache for important data.	Low	High
Cloud platform downtime	If all our server and data are hold in a cloud platform, that is a dangerous decision. If for some reasons, the cloud platforms lose our data even though it may rarely	Backup data services and data replicated redundancy will ensure zero data loss.	Low	High

	happen, our business will suffer.			
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IX. Normal state, Warning state, Alarm state

Normal State	All nodes are functioning and no failure. User participation, bidding, and payments are processed in real-time.
Warning State	Performance decrease with increased latency or unexpected long response times, but the system remains operational. Load balancer or additional processing units are spun up to address the load.
Alarm State	Node failures occur (e.g., load balancer failure, backend server failure, and database failure) potentially impacts user experience or data integrity. The system triggers redundancy mechanisms to maintain operation.

X. Expected Data Growth and Impacts

Expected Data Growth	Impacts
- As the company expands overseas, the number of simultaneous auctions and auction data will grow. - As the company grows and aggressively merging other smaller competitors, the user data grows aggressively as well. The reputation tracking will generate a large volume of data, especially as more users join. - Post-auction video streams will add significant storage demands.	With increasing users and auction, data storage will need to expand. More replication must need. This increases the infrastructure cost and data grid scalability.

XI. Business Continuity Needs

- High availability:** Using distributed cloud infrastructure ensures the auction system 99.5% uptime.
- Disaster recovery:** Using distributed cloud infrastructure ensures the auction system can recover from regional failures.

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3. **Data backup strategy:** Redundancy is must have to avoid data loss when database failure happens.
4. **Data Retention:** Maintain historical data for regulatory requirements and user reputation tracking because the recent lawsuit.
5. **Risk management:** network security for financial transitions, maintain data consistency and prevent data loss. Avoid another law suit.
6. **Communication:** A well document architecture design and runbooks help with the system maintainability.

XII. Architectural Decision Records (ADRs)

ADRs	Context	Status	Decision
Consider Event-driven Architecture	Bidding is an event-based model, event-driven architecture comes in my mind.	Rejected.	Data loss could happen between event propagation, and event-driven architecture is lack of elasticity.
Consider Microkernel Architecture	The company expanding aggressively by merging with smaller competitors, microkernel architecture adopts other systems well.	Rejected.	Microkernel architecture doesn't not scale well, which is not a good fit for auction system.
Use of Space-Based Architecture	Space-Based Architecture supports high performance, high scalability and high levels of elasticity to handle unpredictable spikes in user and request load.	Accepted.	Adopt space-based architecture with distributed processing units for each auction. Supports real-time, isolated, and scalable auction experiences with high consistency.
Distributed Caching Server	Space-Based Architecture mostly relies on replicated caching, but data collisions could happen. Adopt distributed caching instead of replicated caching to ensure data consistency.	Accepted.	Implement a distributed caching server for bidding and auction event to ensure rapid data access and consistency across a large number of concurrent users.

Reference

[1] Mark Richards & Neal Ford, Fundamentals of Software Architecture an Engineering Approach, February 2020, First Edition, CA.

[2] nectar4678, <https://codemia.io/system-design/design-an-auction-system/solutions/sc1tztg/My-Solution-for-Design-an-Auction-system>

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